LUCIDMSK CME PROGRAM

MODULE 03 — POST-TEST

Shoulder MRI: Rotator Cuff, Instability & Arthroplasty

Shoulder · 10 Questions · 1.0 AMA PRA Cat 1 Credit(s)™

Read each question carefully and select the single best answer. A score of ≥ **70%** (7/10 correct) is required to claim CME credit. The complete answer key with detailed explanations begins after Question 10.

POST-TEST QUESTIONS

1 **Parasagittal T1 MRI shows fat content equal to muscle in supraspinatus. Goutallier grade:**

- **A** Grade 1
- **B** Grade 2
- **C** Grade 3
- **D** Grade 4

2 **Supraspinatus stump retracted to the glenoid level. Patte stage:**

- **A** Stage 1
- **B** Stage 2
- **C** Stage 3
- **D** Cannot determine from MRI

3 **Critical Shoulder Angle (CSA) measures 38°. Increased risk for:**

- **A** Glenohumeral OA
- **B** Supraspinatus tear
- **C** Biceps rupture
- **D** AC joint arthritis

4 **Axial MRI shows posterior humeral head subluxation with biconcave glenoid and >2.5 mm step-off. Walch type:**

- **A** A2
- **B** B1
- **C** B2
- **D** B3

5 **rTSA CT shows inferior scapular neck erosion contacting the inferior baseplate screw. Sirveaux notching grade:**

- A Grade 1
- B Grade 2
- C Grade 3
- D Grade 4

6 **Hamada Grade 3 has which ADDITIONAL finding compared to Grade 2?**

- A AHI ≤5 mm
- B GH joint narrowing
- C Acetabularization of the acromion
- D Humeral head collapse

7 **Post-aTSA "rocking horse" sign at the glenoid-cement interface indicates:**

- A Periprosthetic fracture
- B Glenoid component loosening
- C Rotator cuff tear
- D Periprosthetic infection

8 **Which Goutallier grade is a relative contraindication to rotator cuff repair, especially infraspinatus?**

- A Grade 0
- B Grade 1
- C Grade 2
- D Grade 3–4

9 **Seebauer Type IIB rotator cuff arthropathy is characterized by:**

A	Centered, stable; intact CA arch
B	Centered, medialized; early bone loss
C	Decentered; limited escape; acromion erosion
D	Anterior escape; AC joint eroded; most destabilized

10 **A supraspinatus stump remnant of 18 mm is identified after retraction. Surgical significance:**

A	No significance
B	Stump >15 mm predicts repairability (Meyer 2012)
C	Stump <20 mm indicates irreversible atrophy
D	Stump length only relevant for subscapularis

ANSWER KEY & EXPLANATIONS

Q	1	2	3	4	5	6	7	8	9	10
ANS	C	C	B	C	B	C	B	D	D	B

Q1 **Parasagittal T1 MRI shows fat content equal to muscle in supraspinatus. Goutallier grade:** **Answer: C**

A Grade 1
B Grade 2
✓ **Grade 3**
D Grade 4

EXPLANATION Grade 3 = fat equal to muscle. Grade 2 = fat less than muscle. Grade 4 = fat exceeds muscle. Grade ≥2 predicts poorer rotator cuff repair outcomes.

Q2 **Supraspinatus stump retracted to the glenoid level. Patte stage:** **Answer: C**

A Stage 1
B Stage 2
✓ **Stage 3**
D Cannot determine from MRI

EXPLANATION Patte Stage 3 = stump at or beyond glenoid level. Stage 2 = stump at humeral head level. Stage 1 = near greater tuberosity. Stage 3 tears may not be primarily repairable.

Q3 **Critical Shoulder Angle (CSA) measures 38°. Increased risk for:** **Answer: B**

A Glenohumeral OA
✓ **Supraspinatus tear**
C Biceps rupture
D AC joint arthritis

EXPLANATION CSA >35° = increased supraspinatus tear risk — high acromion creates superior shear on the cuff. CSA <30° predicts glenohumeral OA. Normal range = 30–35°.

Q4 **Axial MRI shows posterior humeral head subluxation with biconcave glenoid and >2.5 mm step-off. Walch type:** **Answer: C**

A A2
B B1
✓ **B2**
D B3

EXPLANATION Walch B2 = posterior subluxation + biconcavity + >2.5 mm step-off. B1 lacks biconcavity. B3 = monoconcave posterior wear with >15° retroversion. A2 = centered with central erosion.

Q5 **rTSA CT shows inferior scapular neck erosion contacting the inferior baseplate screw. Sirveaux notching grade:** **Answer: B**

A Grade 1
✓ **Grade 2**
C Grade 3
D Grade 4

EXPLANATION Sirveaux Grade 2 = notch contacts the inferior screw. Grade 1 stays within the pillar. Grade 3 exceeds the screw affecting baseplate stability. Grade 4 extends to the inferior baseplate border.

Q6 **Hamada Grade 3 has which ADDITIONAL finding compared to Grade 2?** **Answer: C**

A AHI ≤5 mm
B GH joint narrowing
✓ **Acetabularization of the acromion**
D Humeral head collapse

EXPLANATION Grade 3 adds acetabularization (concavity/remodeling of the CA arch) to Grade 2's superior migration (AHI ≤5 mm). Grade 4 adds GH narrowing. Grade 5 = head collapse.

Q7 **Post-aTSA "rocking horse" sign at the glenoid-cement interface indicates:** **Answer: B**

A Periprosthetic fracture
✓ **Glenoid component loosening**
C Rotator cuff tear
D Periprosthetic infection

EXPLANATION "Rocking horse" sign = radiolucent lines at the glenoid-cement-bone interface with component tilting — classic sign of glenoid loosening due to failed bone ingrowth or cement mantle failure.

Q8 **Which Goutallier grade is a relative contraindication to rotator cuff repair, especially infraspinatus?** **Answer: D**

A Grade 0
B Grade 1
C Grade 2
✓ **Grade 3–4**

EXPLANATION Goutallier ≥3 (fat ≥ muscle) is a strong contraindication, particularly for infraspinatus. Grade ≥2 infraspinatus atrophy predicts repair failure — rTSA should be considered.

Q9 **Seebauer Type IIB rotator cuff arthropathy is characterized by:** **Answer: D**

- **A** Centered, stable; intact CA arch
- **B** Centered, medialized; early bone loss
- **C** Decentered; limited escape; acromion erosion
- ✓ **Anterior escape; AC joint eroded; most destabilized**

EXPLANATION Seebauer IIB = anterior escape with extensive CA arch, AC joint, and clavicle destruction — most destabilized pattern, highest rTSA complication risk. Type IA = centered and stable (most favorable).

Q10 **A supraspinatus stump remnant of 18 mm is identified after retraction. Surgical significance:** **Answer: B**

- **A** No significance
- ✓ **Stump >15 mm predicts repairability (Meyer 2012)**
- **C** Stump <20 mm indicates irreversible atrophy
- **D** Stump length only relevant for subscapularis

EXPLANATION Meyer DC et al. (2012): remnant stump >15 mm predicts successful repair and tissue hold. Report stump length with Patte stage and Goutallier grade together for complete surgical assessment.